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# Exploring the Late Medieval Adriatic Wax Trade with Neo4j

Vladimir Aleksic\*<sup>1</sup>

<sup>1</sup>Free University Berlin (FU Berlin) – Germany

## Abstract

Studies of medieval social and economic history based on double-entry ledgers, the earliest examples of which date back to the second half of the 13th century in Italy, each of them bearing unique features, require specific methodological approaches. This type of accounting required records concerning individual business activities to be entered into the Reminder, Diary, and Main Book. Furthermore, the latter tracked both the debit side (*dare* – money going out) and the credit side (*avere* – money coming in) of each transaction to maintain a clear overview of the balance between income and expenses. Similar adaptations of the standard accounting model to specific commercial needs can also be observed in three manuscripts attributed to the prominent Ragusian (Dubrovnik) merchants Luca and Nicola C(h)aboga (Kabužić). This source offers valuable insight into the large-scale trade of wax and precious metals between the central Balkans and the Adriatic economic centres of Italy. The DFG-funded project "Graph Database Analysis of the Trans-Adriatic Trade Records in the Kabužić Brothers' Ledgers (1426–1433)" is based on the critical edition (Kovačević-Kojić 1999). Our focus is on the company's activities in purifying raw wax and sourcing refined wax from the Balkan hinterland, which gave rise to an extensive network of sellers, buyers and ship owners.

The primary research question of our presentation concerns the correct interpretation of the complex double-entry structure, which contains temporal, numerical (currency-related), textual, geographical, and other types of data. The complexity of this challenge increases further when graph databases are used to investigate the business strategies of the Caboga brothers within the broader economic, historical, geographical, cultural and political context on the edges of two worlds - Italian and the Central Balkans. This includes transport difficulties related to caravan and maritime communication, trade bans issued by the Emperor/King Sigismund of Luxembourg, the use of different currencies and exchange rates, and many other factors. For example, the frequent inconsistencies in data structure and variations in personal names-common in many medieval sources and also evident in the Caboga ledgers-make the design of optimal data and instance models particularly challenging.

This is where Neo4j (desktop version) becomes especially useful, as graph databases store information in the form of nodes (e.g. :Wax) and relationships (e.g. -(BOUGHT)→) between nodes, as well as key-value pairs (e.g. {fName: 'Luca', lName: ('Caboga', 'Chaboga')}). This feature allows for a greater degree of flexibility in modelling complex historical data and tracing irregularities and exceptions within the sources (Trogitz 2019). By using the

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\*Speaker

Cypher query language, the data is traversed to address questions about the technical aspects of the wax purification process - Capture nu. 1, as well as the volume, temporal and geographical distribution of the wax trade, and fluctuations/patterns in the associated social network -Capture nu. 2 (Kuczera 2019).

The technical aspects of converting textual data into CSV files will also be addressed, as several methods have been applied - including the CATMA annotation tool, manual data extraction, and Google NotebookLM - each producing different results.

## References

Kovačević-Kojić, Desanka. *Trgovačke knjige braće Kabužić (Caboga) 1426–1433: Account Books of the Caboga (Kabužić) Brothers 1426–1433. Spomenik (Srpska Akademija Nauka i Umetnosti Beograd)* 137. Beograd, 1999.

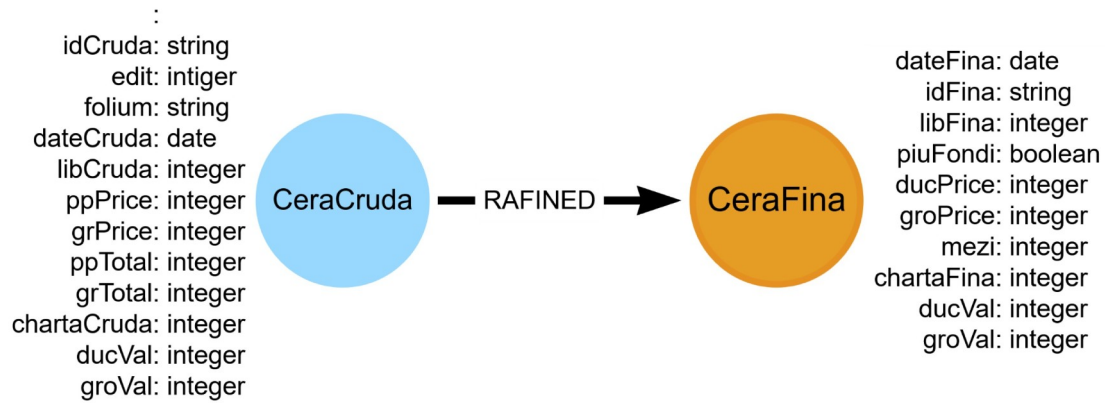
Kuczera, Andreas. 2019. "Das Nürnberger Briefeingangsregister als Quelle für die Informationsverarbeitung in der Stadt des 15. Jahrhunderts." In *Informationsverarbeitung in der Stadt des 12.–16. Jahrhunderts*, edited by Eric Burkart and Vincenz Schwab, 56–71. Trier: Trierer Zentrum für Mediävistik, 2019.

Trognitz, Martina. 2019. "A Graph Database of Aegean Seals with Uncertain Attributes." In *Die Modellierung des Zweifels – Schlüsselideen und -konzepte zur graphbasierten Modellierung von Unsicherheiten, Sonderbände der Zeitschrift für digitale Geisteswissenschaften* 4, edited by Andreas Kuczera, Thorsten Wübbena, and Thomas Kollatz, 15–(end page if known). Wolfenbüttel: 2019.

**Keywords:** Historical Ledgers, Economic History, Adriatic Basin, Early 15th Century, Graph Technologies

**Capture nu. 1**

**A data model describing the purification process**

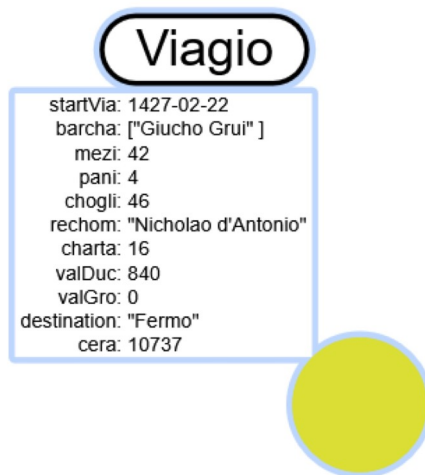


**A CYPHER query to calculate the temporal difference, in days, between the start and end of the purification process.**

```
MATCH (c:CeraCruda)-[:RAFINED]->(f:CeraFina)  
RETURN duration.between(c.dateCruda, f.dateFina).days AS date_diff;
```

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Capture nu. 2  
An instance model of the Viagio node



The data stored describes the starting day of the trip to Fermo, the name of the person to whom the shipment was dedicated (i.e., the shipowner), as well as the value in ducats and grossi, amount (10737 pounds), and method of wax packaging.

A query to find all starting dates of shipment which involved specific person in specific place

```
MATCH (v:Viagio)
WHERE v.rechon = "Nicholao d'Antonio"
AND v.destination = "Fermo"
RETURN v.startVia AS Trip_starts;
```